Pre-Sales is the stage before a company sells a software or IT solution to a customer.

Process:

- Understanding the client's problem
- Explaining how your product/solution can solve it
- Demonstrating technical capability
- Helping close the deal

Scenario example: Client wants: AI-enabled Investor's News Recommendation System.

First step is not coding. It is discovery and problem definition.

Understand Business Problem First Like:

Why do you need this?
What business outcome?
What decisions will it improve?
How are they doing it today? Tools used?
Pain points exist?

Scenario should be in mind like

Might be struggling with segmented investors, news overloading without filtering, time spend in manual news analysing.

Gather Requirements / Discovery Questions

Investor's Types :

How do you classify investors? (Conservative, Aggressive, Balanced etc.)

Data availability?

Automation level? (Manual or automated process)

News Recommendation:

- What news sources?
- How do you match news to portfolios?
- Real-time or delayed news?
- Any filter?
- You use Personalized or generic?

Data

- Where is data stored?
- Format?
- Access permissions?
- Is data structured or unstructured?

AI Expectations

- Explainable AI?
- Governance?
- Data privacy?
- Hosting preference (Cloud / Data centre)?

Suppose got an answers from client:

- Manual investor tagging
- Spreadsheets
- News read manually
- No personalization

Discovery Document

Executive Summary

The client requires an AI-enabled investor persona segmentation and news recommendation system to:

- Personalize investor insights
- Improve news relevance
- Enhance client engagement
- Support investment decision making

The solution will use AI/ML techniques to classify investor personas and match news with investment preferences.

Client problem → AI solution → Business value

Business Objectives

- Automated investor segmentation
- Personalized news recommendations
- Faster insight delivery
- Reduced manual effort
- Improved client satisfaction

Success is measured by:

- News Recommendation relevance
- Engagement metrics
- User feedback
- Operational efficiency

Current State

- Investor classification is manual
- News is consumed generically
- No AI-driven personalization
- Data exists but not leveraged fully
- Insights are delayed

Pain points:

- Time-consuming analysis
- Limited personalization
- Inconsistent recommendations

Proposed Solution

AI-driven system with:

- Persona segmentation (LLM + ML)
- News recommendation engine
- Hybrid architecture
- Synthetic data for demo
- Azure-based deployment (if enterprise)

We will use a hybrid AI approach combining LLM for reasoning and ML for classification. Personas will be derived from behavioural attributes, and news recommendations will be matched using relevance scoring.

Components:

- Frontend (React)
- Backend (Python)
- AI Layer (LLM)
- ML Layer (XGBoost / Scikit-learn)
- Data Layer (Structured storage (SQL/NoSQL))

Assumptions

- Client data availability
- Data access permissions
- AI usage permitted
- Demo can use synthetic data
- Cloud deployment allowed (if required)

Constraints

- Data privacy
- Regulatory compliance
- Budget
- Timeline
- Explainability requirements

Success Criteria

- Accurate investor segmentation
- Relevant news recommendations
- Improved engagement
- Reduced manual effort

Risks

- Data quality issues
- Model bias / Model accuracy
- Integration challenges
- Governance requirements

Mitigation:

- Validation
- Monitoring
- Explainable AI (collection of procedures and techniques that enable machine learning algorithms to produce output)

Basic Architecture

User → Frontend → Backend → AI/ML → Recommendation Engine → Output

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Solution Architecture (for Development and Production)

Solution Architecture (for Development and Production)

Technologies:

Frontend -	React Js (For User dashboard)
Backend -	Python (FastAPI)
AI -	Hybrid (LLM + ML)
LLM -	Ollama
ML -	XGBoost / Scikit-learn
Data -	Test data
Storage -	JSON/CSV
Text to Speech -	gTTS (optional)

Advantages:

Offline
No cloud cost
Fast iteration
Demo friendly

C4 Architecture Diagram

C4 – Level -1 - Context

User → Investor Recommendation System → Local LLM – Ollama → Local Database

C4 – Level -2 – Container diagram

React Frontend → FastAPI Backend → AI Layer (Agents + ML) → Ollama Runtime → MongoDB / JSON Storage

C4 – Level -3 - Componenet diagram (Backend)

Fast - API

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Authentication Controller

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News Controller

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AI Orchestrator

├── Persona Agent (LLM)

├── News Classifier (ML)

├── Recommendation Engine

└── Conversation Agent (LLM)

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Prompt Manager (JSON Templates)

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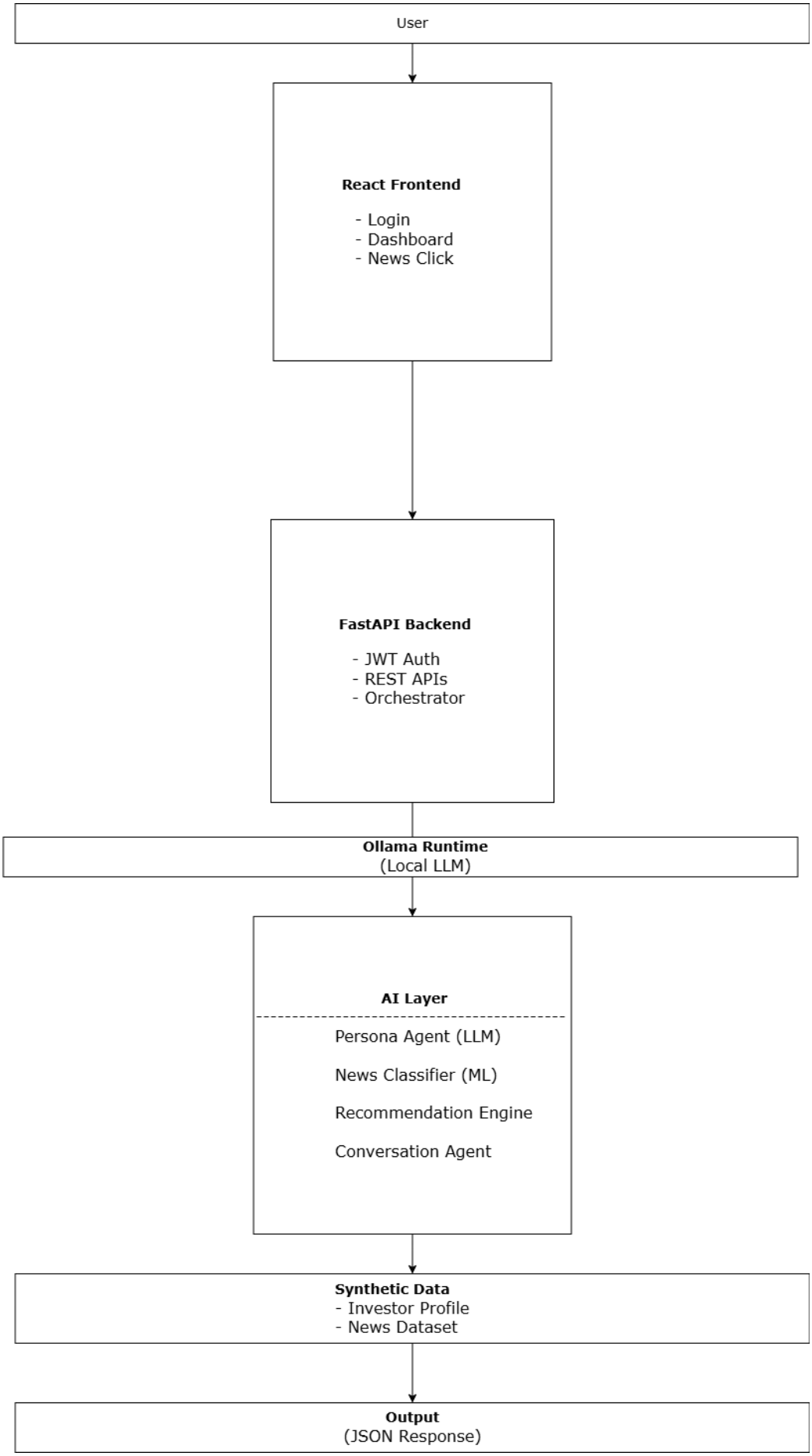
Data Access Layer

Full Technical Block Diagram (Application + Data Flow)


User
React Frontend
- Login
- Dashboard
- News Click
FastAPI Backend
- JWT Auth
- REST APIs
- Orchestrator
Ollama Runtime
(Local LLM)
AI Layer
Persona Agent (LLM)
News Classifier (ML)
Recommendation Engine
Conversation Agent
Synthetic Data
- Investor Profile
- News Dataset
Output
(JSON Response)